

Mexiletine and Local Anaesthetic Binding Sites on Nav Channels

A Structured Literature Review with Relevance to the Nav1.5 N347K Brugada System

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1. Scope and Search Strategy

This review consolidates the experimental and computational literature on where mexiletine and structurally related local anaesthetic (LA) / class-Ib antiarrhythmic drugs bind on voltage-gated sodium (Nav) channels, with emphasis on the cardiac isoform Nav1.5. Its purpose is to provide an external reference frame for the binding sites identified in our own N347K computational study.

The search was conducted across PubMed, Google Scholar and journal databases using three layers of queries: (i) mexiletine-specific terms across all Nav isoforms (e.g. mexiletine binding site, mexiletine local anaesthetic receptor, mexiletine cryo-EM); (ii) Nav1.5-specific terms for LA / antiarrhythmic binding (e.g. Nav1.5 local anaesthetic site, flecainide / lidocaine Nav1.5 structure); and (iii) mechanistic and structural reference terms (Nav pore blocker DIV-S6, fenestration drug binding, state-dependent block). Only primary structural, electrophysiological, mutagenesis and molecular-modelling studies were retained.

2. The Canonical Local Anaesthetic Receptor Site (DIII-S6 / DIV-S6)

The classical LA receptor is formed inside the central pore cavity by residues contributed by the S6 helices of multiple domains, principally Domain III S6 and Domain IV S6. The aromatic residue at the DIV-S6 position — F1760 in human Nav1.5 — is the single most important determinant of both tonic and use-dependent block. Mutation of this residue to alanine (F1760A) markedly attenuates drug binding, and this position is shared by mexiletine, lidocaine and several other pore blockers [1], [2].

Nakajima and colleagues compared state-dependent mexiletine affinities across Nav1.2, Nav1.4 and Nav1.5 and systematically mutated DIV-S6 residues [1]. The largest affinity change resulted from alanine substitution of the DIV-S6 phenylalanine, and high-affinity binding in the open state of Nav1.4 and Nav1.5 was attributed to a π - π interaction between the drug's aromatic ring and this phenylalanine [1]. This is direct experimental support for the role of an aromatic DIV-S6 residue, the equivalent of which (PHE1760) emerges as the top-ranked pocket determinant in our MD-derived N347K structures.

An independent line of evidence comes from a study of bisphenol A (BPA) block of Nav1.5 [2]. By comparing wild-type and F1760A channels, the authors demonstrated that BPA and mexiletine share an overlapping binding site delimited by F1760 and contiguous with the DIII-IV pore fenestration, and steered MD indicated the DIII-IV fenestration as a viable access pathway [2]. The convergence of the F1760 pore site and a DIII-IV fenestration access route in that work parallels the two-site picture (DIV-S6 pore site plus DIII/IV junction) seen in our own analysis.

Both mexiletine and lidocaine carry an ionizable amino group and an aromatic ring, recognised as the two pharmacophoric moieties that engage, respectively, in cation- π / electrostatic and hydrophobic interactions with the conserved S6 aromatic residues lining the inner vestibule [3]. The receptor is highly conserved across Nav isoforms, which is consistent with the low subtype selectivity of these drugs [3].

3. Structural (Cryo-EM) Evidence for Pore-Cavity Binding in Nav1.5

Direct structural confirmation of pore-cavity binding in the cardiac channel comes from the cryo-EM structure of flecainide bound to rat Nav1.5 (PDB 6UZ3) [4]. Flecainide, quinidine and propafenone are all accommodated within the central cavity of the Nav1.5 pore domain, although with distinct, non-overlapping poses [5]. This establishes the central cavity beneath the selectivity filter as the principal structurally-resolved drug-binding region in Nav1.5.

A computational comparison of flecainide binding in Nav1.5 and the atrial channel Kv1.5, benchmarked against the 6UZ3 structure, identified a common central-cavity binding pattern composed of a hydrophobic patch and a polar region contributed by S6 helices and the P-loop [6]. The Nav1.5 contact residues reported there include a DIV-S6 phenylalanine (numbered F1762 in that study) together with DIII residues such as F1420, I1456/I1457, F1461 and I1464 [6]. The close correspondence between this DIV-S6 aromatic residue and the F1760 determinant from mutagenesis studies illustrates a recurring (and numbering-sensitive) theme: an aromatic DIV-S6 residue anchors the drug in the pore cavity.

Broader cryo-EM mapping of Nav-channel drug sites — most comprehensively on Nav1.7 — has revealed at least three classes of site: a site beneath the intracellular activation gate, the central-cavity / selectivity-filter region, and the lateral fenestrations between pore-domain helices [5]. State-dependent and smaller hydrophobic drugs frequently occupy the fenestrations, which lie at interfaces between pore-domain helices and are otherwise filled by membrane phospholipid tails when unoccupied [7]. Mixed binding modes, in which a drug populates both a fenestration and the central pore, have been described for lacosamide and mexiletine [7].

4. Protonation State and Molecular-Dynamics Perspectives

Mexiletine ($pK_a \approx 9.1$) is predominantly protonated (+1) at physiological pH, and its charge state directly affects both where it binds and how well computational methods reproduce experimental structures. Monte-Carlo docking of mexiletine into a cryo-EM Nav1.5 structure yielded two pose ensembles in contact with known MEX-sensing residues of S6(III), S6(IV) and P1(IV): in both, the ligand ammonium group approached the electronegative outer-pore site formed by backbone carbonyls, while the aromatic ring extended either along the inner pore (a use-dependent-block-like pose) or into the III/IV fenestration (a tonic-block-like pose) [8].

The importance of protonation is reinforced by a perspective on Nav1.5-blocker simulations, which noted that for flecainide the protonated form adhered more closely to the experimental cryo-EM pose than the deprotonated form, supporting the view that the structurally-observed drug is protonated [7]. This is methodologically aligned with our decision to model the physiologically dominant protonated mexiletine, and with our finding that the protonated drug is electrostatically steered toward the acidic Domain III/IV junction rather than remaining in the neutral-form P-loop pose.

It is equally important to note where simulations and structures diverge. The same perspective cautions that metadynamics-predicted sites do not always match later cryo-EM structures — for example the riluzole/lamotrigine sites on Nav1.7 [7]. This is a useful caveat for interpreting our docking results, where the rigid-receptor underestimation of the electrostatically-driven Domain III/IV site is itself an example of method-dependent disagreement.

5. Comparative Map of Reported Binding Sites

Table 1 summarises the principal binding sites reported for mexiletine and related Nav blockers, the method used to identify each, and the corresponding region of the channel. The final column relates each entry to the sites identified in our N347K study.

Drug / ligand	Site & channel region	Channel	Method	Relation to our N347K sites
Mexiletine	DIV-S6 aromatic (F1760), central pore; π - π with Phe	Nav1.5 (also 1.2/1.4)	Mutagenesis + electrophysiology + homology model [1]	Matches our Site 1 (DIV-S6 / PHE1760), primary druggable pocket
Mexiletine / BPA	F1760 pore site, contiguous with DIII-IV fenestration	Nav1.5	F1760A mutant + patch clamp + steered MD [2]	Supports both Site 1 (F1760) and the DIII/IV access route
Mexiletine / Thio-Me2	S6(III)/S6(IV)/P1(IV); inner-pore (UDB) or III/IV fenestration (TB)	Nav1.5	Monte-Carlo docking into cryo-EM [8]	Inner-pore vs III/IV duality mirrors our two-site model
Flecainide	Central cavity below selectivity filter	Nav1.5 (PDB 6UZ3)	Cryo-EM [4]; MD comparison [6]	Confirms central-pore cavity adjacent to our Site 1
Flecainide / quinidine / propafenone	Central cavity, non-overlapping poses	Nav1.5	Cryo-EM [5]	Defines pore cavity as main resolved drug region
Lacosamide / mexiletine	Mixed: fenestration + central pore	Nav1.5 / Nav1.7	Cryo-EM + simulation review [7]	Fenestration occupancy relevant to DIII/IV junction (Site 2)
Carbamazepine / bupivacaine / lacosamide	Site beneath intracellular gate (site BIG)	Nav1.7	Cryo-EM [5]	Distinct gate site; not our primary focus

Table 1. Reported binding sites for mexiletine and related Nav blockers, mapped to the two sites identified in the N347K study.

6. Mexiletine Rescue of N347K and Related SCN5A Brugada Variants

The clinical and electrophysiological starting point for our project is the work of Frosio et al., who characterised three novel SCN5A Brugada variants — p.A344S, p.N347K and p.D349N — located in the Domain I P-loop region [9]. Automated patch-clamp recordings showed that N347K and D349N essentially abolished inward sodium current, yet the channels remained present at the cell membrane, indicating a dysfunctional rather than fully trafficking-impaired channel [9]. Mexiletine treatment recovered current density in N347K and D349N, whereas A344S responded only weakly [9]. This mutation-selective rescue — full current recovery for N347K but a markedly different ($\approx 85\%$ peak-current-reducing) response for the neighbouring A344S — is precisely the selectivity our structural work seeks to explain.

Mexiletine-mediated rescue of loss-of-function SCN5A variants is a recurring observation beyond this single study. Pore-region SCN5A mutations causing trafficking defects have repeatedly been shown to be partially rescued by mexiletine incubation, with restored peak I_{Na} attributed in several cases to increased cell-surface expression of the channel [10], [11]. Whether rescue operates through a pharmacological-chaperone / trafficking mechanism, through a direct conformational effect at the pore, or through an allosteric route remains variant-dependent and incompletely resolved — a gap our N347K electrostatic-remodelling hypothesis directly addresses.

7. Synthesis and Identified Gaps

Three conclusions emerge from the literature that frame our own findings. First, the DIV-S6 aromatic residue (F1760 in Nav1.5) is the best-established single determinant of mexiletine binding, confirmed by mutagenesis, electrophysiology and structural studies [1]–[6]; this externally validates our identification of PHE1760 as the primary druggable pocket. Second, the pore cavity, the DIII/IV fenestration and the intracellular-gate region constitute distinct, experimentally-supported binding regions, and protonated class-Ib drugs can populate more than one of these — supporting our two-site model rather than a single-site picture [5], [7], [8]. Third, the protonation state materially affects both binding location and the agreement between simulation and structure [7], [8], reinforcing the methodological importance of modelling the physiologically dominant protonated form.

The principal gap in the literature is mechanistic: although mexiletine rescue of N347K and similar Brugada variants is well documented electrophysiologically [9]–[11], there is no structural or dynamical explanation linking the drug's binding location to the restoration of sodium current in a specific mutant. To our knowledge, no prior study has examined how the K347+ charge reshapes the electrostatic landscape of Nav1.5 or how this directs a protonated drug to the Domain III/IV junction. This is the specific contribution of the present project, and the holoWT protonated simulation now in progress will provide the wild-type comparison ($\Delta\Delta G$) needed to test whether the Domain III/IV preference is genuinely N347K-selective.

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